Franck Portier
2016-2017

M1 Macro

Group A

Session 4

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Endowment economy

y_1 , y_2 is stochastic

r is given and deterministic

$$\max \log c_1 + \beta E_1 \log c_2$$

s.t.

$$\begin{cases} c_1 + S = y_1 \\ c_2 = y_2 + (1+r) S \end{cases}$$

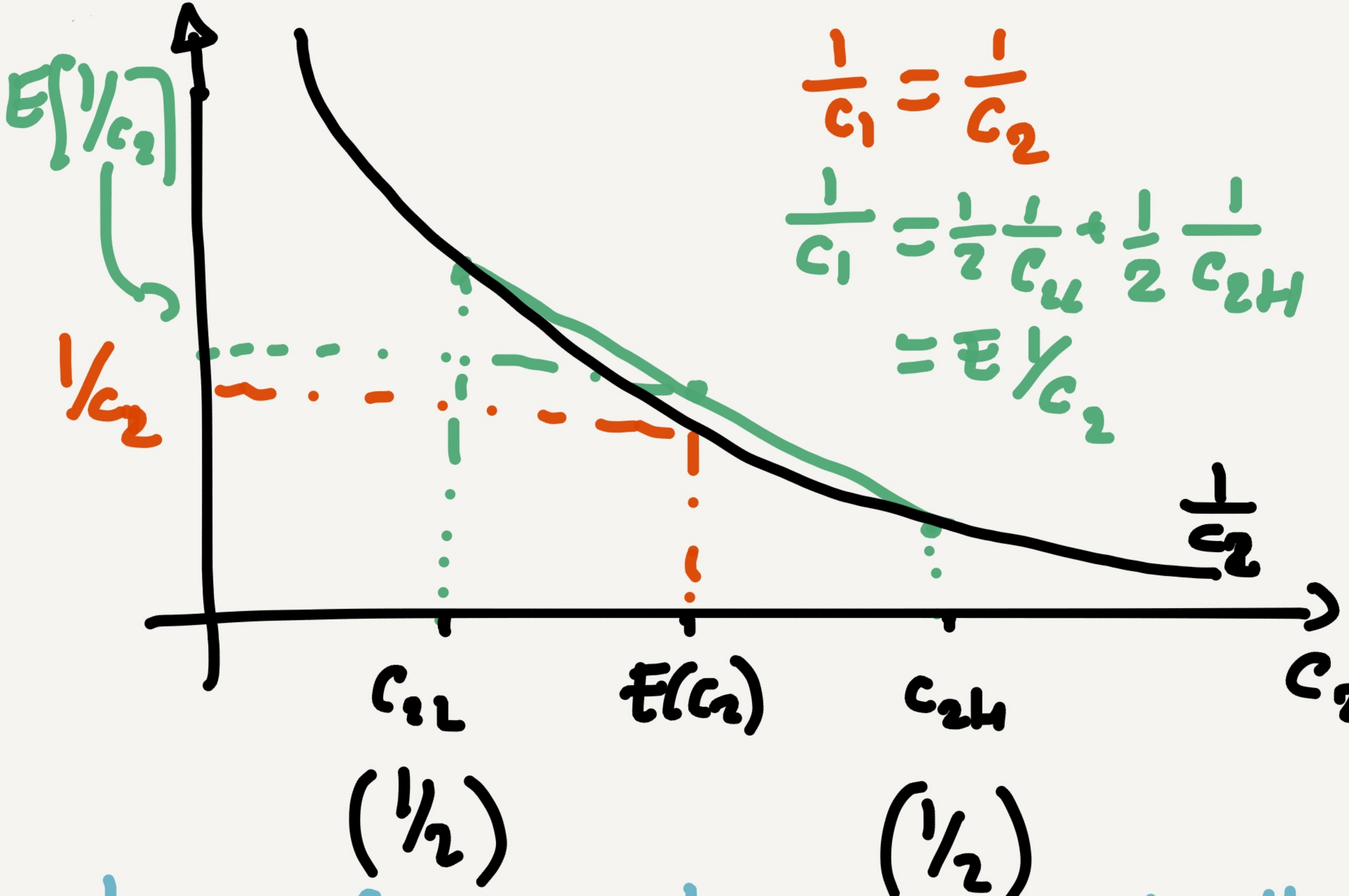
With uncertainty

$$\frac{1}{c_1} = \beta(1+r) E_1\left(\frac{1}{c_2}\right)$$

Without uncertainty $\quad y_2 \rightarrow E[y_2]$

$$\frac{1}{c_1} = \beta(1+r) \frac{1}{E[c_2]}$$

<u>Assumption</u> $\quad \beta = \frac{1}{1+r} \Rightarrow \beta(1+r) = 1$



$$\frac{1}{c_1} = \frac{1}{c_2}$$

$$\frac{1}{c_1} = \frac{1}{2}\frac{1}{c_{2L}} + \frac{1}{2}\frac{1}{c_{2H}}$$

$$= E\, 1/c_2$$

$\frac{1}{c_2} < E[1/c_2] \Rightarrow \frac{1}{c_1}$ with constants $< \frac{1}{c_1}$ with linear.

$\frac{1}{c_1}$ with certainty $<$ $\frac{1}{c_1}$ with uncertainty

$\Rightarrow$ c_1 with certainty $>$ c_1 with uncertainty

S ______ $<$ S ______

PRECAUTIONARY SAVING

Labour supply

$$\max \; \log c_1 + \beta \log c_2 + \log(1-N_1) + \beta \log(1-N_2)$$

s.t. $$\begin{cases} c_1 + S = W_1 N_1 \\ c_2 = W_2 N_2 + (1+r)S \end{cases} \rightarrow c_1 + \frac{c_2}{R} = W_1 N_1 + \frac{W_2 N_2}{R}$$

Choose optimally $\begin{cases} c_1, c_2, S \\ N_1, N_2 \end{cases}$

W_1, W_2, r are given

$1+r = R$

$$\mathcal{L} = \log c_1 + \beta \log c_2 + \log(1-N_1) + \beta \log(1-N_2)$$
$$+\lambda\left(W_1 N_1 + \frac{W_2 N_2}{R} - c_1 - \frac{c_2}{R}\right)$$

<u>FOC</u>

$/c_1: \quad \frac{1}{c_1} = \lambda \quad (1)$

$/c_2: \quad \frac{\beta}{c_2} = \lambda / R \quad (2)$

$/N_1: \quad \frac{1}{1-N_1} = \lambda W_1 \quad (3)$

$/N_2: \quad \frac{\beta}{1-N_2} = \lambda \frac{W_2}{R} \quad (4)$

$c_1\ c_2\ N_1\ N_2$: 4 unk. $+\lambda$

<u>eq</u>: 5 eq

$$+\lambda\left(W_1 N_1 + \frac{W_2 N_2}{R} - c_1 - \frac{c_2}{R}\right) = 0, \ \lambda \geq 0$$

Solution

Assume again $R = \frac{1}{\beta}$

(1) and (2) $\Rightarrow C_1 = C_2 = C$

(3) $\frac{1}{1-N_1} = \lambda w_1 \Leftrightarrow 1 - N_1 = \frac{c}{w_1}$

$$\Leftrightarrow N_1 = 1 - \frac{c}{w_1}$$

(4) $$N_2 = 1 - \frac{c}{w_2}$$

1Bc

$$C + \frac{C}{R} = W_1\left[1 - \frac{C}{W_1}\right] + \frac{W_2}{R}\left[1 - \frac{C}{W_2}\right]$$

$$C\left[1 + \frac{1}{R} + 1 + \frac{1}{R}\right] = W_1 + \frac{W_2}{R}$$

$$2C\underbrace{\left[1 + \frac{1}{R}\right]}_{\frac{1+R}{R}} = W_1 + \frac{W_2}{R}$$

$$\boxed{C = \frac{1}{2}\,\frac{R}{1+R}\left[W_1 + \frac{W_2}{R}\right]}$$